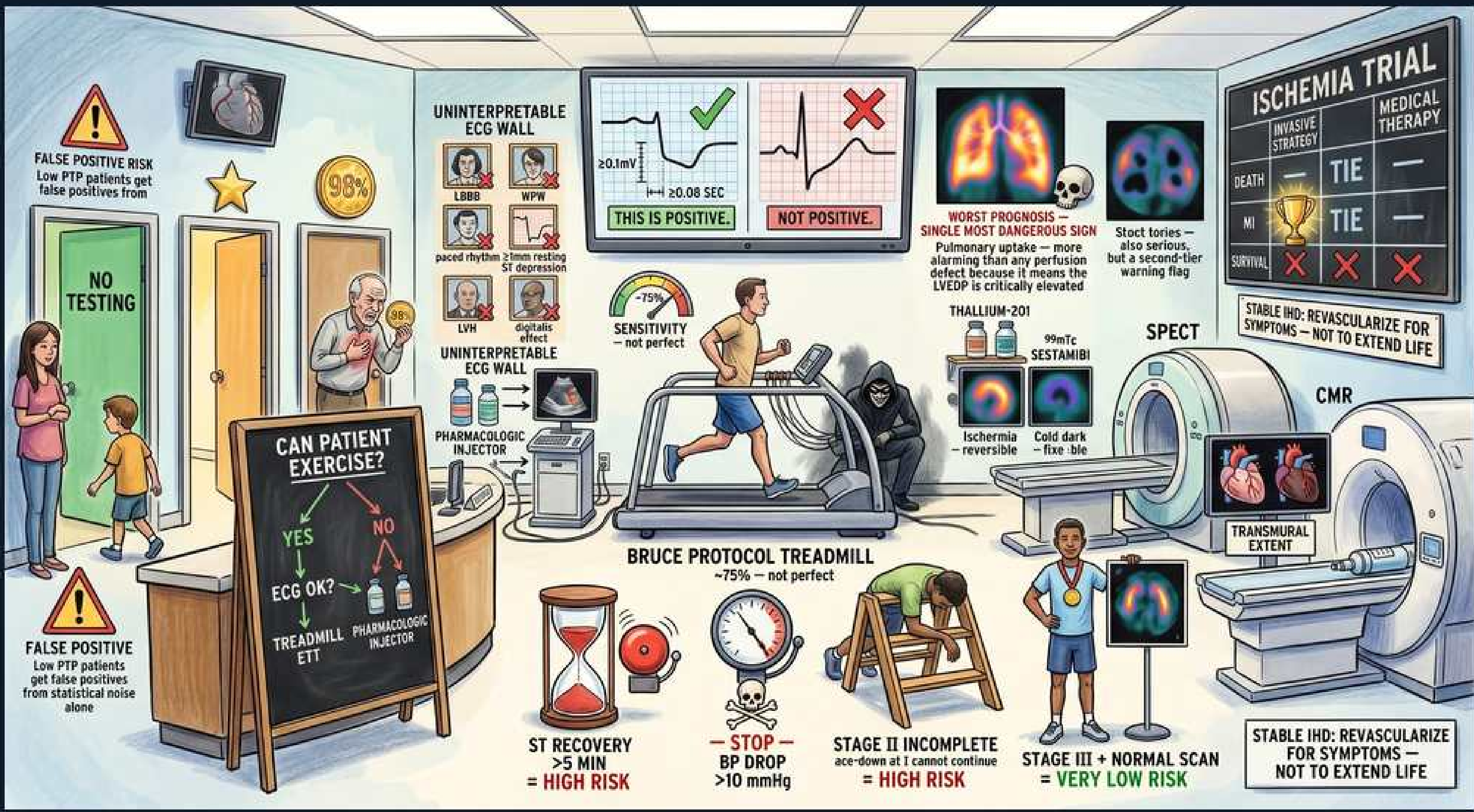
IHD — Exercise Stress Test: Interpretation & High-Risk Findings



1. Sensitivity ~75%

WHAT YOU SEE

Gauge top-left '~75% SENSITIVITY — not perfect'

WHAT IT ENCODES

Exercise ECG has only ~68–75% sensitivity and ~70–80% specificity for obstructive CAD, so a normal test does not fully exclude significant disease. Sensitivity climbs when the patient reaches ≥85% of age-predicted maximum heart rate (220 − age) — a submaximal test (failure to reach target HR) is non-diagnostic, not reassuring. Pretest probability still governs interpretation (Bayes): a negative test in a high-risk patient may still be a false negative.

BOARD TRAP

WRONG: a negative exercise stress test rules out significant CAD. Discriminator: with only ~75% sensitivity, a normal result cannot exclude CAD — especially in high pretest-probability patients or a submaximal study (HR <85% predicted), where further testing is warranted.

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2. Positive = ≥1 mm ST dep

WHAT YOU SEE

ECG strips 'FLAT/DOWNSLOPING ≥0.1 mV, ≥0.08 sec — THIS IS POSITIVE'

WHAT IT ENCODES

A positive exercise ECG = ≥1 mm (0.1 mV) of HORIZONTAL or DOWNSLOPING ST depression measured 80 ms (0.08 s) after the J point, in ≥1 lead. The depression reflects subendocardial ischemia. ST ELEVATION in a non-infarct lead is an even more ominous (and rarer) marker localizing transmural ischemia.

BOARD TRAP

WRONG: upsloping ST depression counts as a positive result. Discriminator: only flat/horizontal or downsloping ≥1 mm ST depression is diagnostic — upsloping depression is non-diagnostic/normal, and calling it positive overreads the test.

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3. Upsloping ≠ positive

WHAT YOU SEE

'UPSLOPING ≠ POSITIVE' with red X

WHAT IT ENCODES

Upsloping ST depression is a normal or non-diagnostic exercise response and should NOT be labeled ischemia. Only horizontal or downsloping ST depression ≥1 mm meets criteria. Treating upsloping changes as a positive result generates false positives and unnecessary downstream angiography.

BOARD TRAP

WRONG: upsloping ST depression is a positive (ischemic) finding. Discriminator: the morphology matters — upsloping = benign/non-diagnostic; horizontal or downsloping = positive. This morphology distinction is the single most common interpretation error tested.

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4. Bruce protocol treadmill

WHAT YOU SEE

Man running, 'BRUCE PROTOCOL' on treadmill

WHAT IT ENCODES

The standard Bruce protocol is a graded treadmill test increasing speed and incline every 3 minutes; the goal is to reach ≥85% of age-predicted max HR. Workload is reported in METs, and exercise capacity (METs achieved) is itself a powerful prognostic marker — <5 METs portends a worse prognosis. The Duke treadmill score combines exercise time, ST deviation, and angina to risk-stratify.

BOARD TRAP

WRONG: stopping the test as soon as the patient is mildly tired (before adequate HR/workload) still yields a valid negative. Discriminator: a submaximal study that never reaches 85% predicted HR is non-diagnostic, not reassuring — low METs/early termination is itself a high-risk sign, not a normal test.

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5. ST recovery >5 min = high risk

WHAT YOU SEE

Alarm bell 'ST RECOVERY >5 min = HIGH RISK', hourglass

WHAT IT ENCODES

ST depression that persists more than 5 minutes into the recovery phase is a HIGH-RISK finding suggesting severe/multivessel disease — prolonged ischemia that is slow to resolve. Other high-risk markers: ST depression ≥2 mm, depression at low workload (<5 METs / early in stage I–II), ST elevation, and exertional hypotension. High-risk results warrant invasive coronary angiography.

BOARD TRAP

WRONG: ST changes that linger into recovery are reassuring because the patient 'cooled down.' Discriminator: prolonged recovery-phase ST depression (>5 min) signals more severe ischemia, not resolution — it triggers angiography, not reassurance.

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6. BP drop >10 = high risk / stop

WHAT YOU SEE

'BP DROP >10 mmHg — STOP — HIGH RISK', pressure gauge falling

WHAT IT ENCODES

A FALL in systolic BP with exercise (exertional hypotension, drop ≥10 mmHg below baseline or failure to rise) signals severe ischemia or LV systolic dysfunction / left main or 3-vessel disease — it is a high-risk finding and an indication to STOP the test. Normally systolic BP rises with exertion; a drop means the failing left ventricle cannot augment cardiac output.

BOARD TRAP

WRONG: a drop in blood pressure during exercise is a benign vasovagal response. Discriminator: exertional hypotension reflects severe LV dysfunction / critical (left main, 3-vessel) disease — it is a danger sign mandating test termination, not a normal finding.

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7. Stage II incomplete = high risk

WHAT YOU SEE

Collapsed runner 'STAGE II INCOMPLETE — can't continue at 1 — HIGH RISK'

WHAT IT ENCODES

Low exercise capacity — inability to complete early stages, early termination, or achieving <5 METs — is itself a strong high-risk/poor-prognosis marker independent of ST changes. Functional capacity is one of the most powerful predictors of cardiovascular outcomes; a patient who fails at low workload needs further evaluation even if the ECG is unrevealing.

BOARD TRAP

WRONG: as long as no ST changes occurred, an aborted low-workload test is negative/reassuring. Discriminator: poor functional capacity (early termination, low METs) is a high-risk finding on its own — the absence of ST changes does not make a low-workload test reassuring.

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8. Large/multiple perfusion defects

WHAT YOU SEE

Top-right nuclear perfusion images showing several dark (under-perfused) patches across the heart — large/multiple reversible defects = extensive multivessel ischemia, a high-risk picture.

WHAT IT ENCODES

On stress perfusion imaging, large or multiple REVERSIBLE perfusion defects indicate extensive jeopardized myocardium (multivessel ischemia) and are high-risk, favoring angiography ± revascularization. A reversible defect (present on stress, fills in at rest) = inducible ischemia; a fixed defect (absent on both) = prior scar/infarct.

BOARD TRAP

WRONG: a single small reversible defect and large multivessel defects carry the same prognosis. Discriminator: defect SIZE and NUMBER drive risk — large/multiple reversible defects mean extensive ischemia and high risk, unlike an isolated small defect.

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9. Pulmonary uptake = worst sign

WHAT YOU SEE

Skull, 'PULMONARY UPTAKE → SINGLE MOST DANGEROUS PROGNOSTIC SIGN — WORST PROGNOSIS'

WHAT IT ENCODES

Increased LUNG uptake of radiotracer on nuclear (thallium) imaging reflects exercise-induced LV dysfunction with elevated pulmonary capillary pressure — it is among the most ominous prognostic findings, signaling severe/extensive (often left main or 3-vessel) disease. Transient ischemic dilation (TID) of the LV cavity on stress images is a similarly high-risk marker of balanced multivessel ischemia.

BOARD TRAP

WRONG: pulmonary tracer uptake is an attenuation artifact / soft-tissue overlap and can be ignored. Discriminator: increased lung uptake is a real, high-risk prognostic sign of LV dysfunction — never dismiss it as artifact.

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10. Stage III + normal = very low risk

WHAT YOU SEE

Medal, 'STAGE III + NORMAL SCAN = VERY LOW RISK'

WHAT IT ENCODES

A patient who completes a high workload (e.g., Bruce stage III–IV, ≥10 METs) with a NORMAL perfusion scan and no ST changes has an excellent prognosis — annual cardiac event rate <1%. Good functional capacity with a normal study is among the most reassuring results and generally needs no further testing.

BOARD TRAP

WRONG: even a normal high-workload study requires confirmatory invasive angiography. Discriminator: completing a high workload with a normal scan confers very low risk (<1%/yr) — invasive angiography is not indicated and would add only procedural risk and false positives.

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11. SPECT (thallium/sestamibi)

WHAT YOU SEE

Large doughnut SPECT gamma-camera scanner (thallium-201 / Tc-99m sestamibi) — the imaging ring that maps perfusion, lighting reversible ischemia versus dark fixed scar.

WHAT IT ENCODES

SPECT myocardial perfusion imaging (thallium-201 or Tc-99m sestamibi/tetrofosmin) localizes and quantifies ischemia by comparing stress and rest images. A REVERSIBLE defect (perfusion deficit on stress that normalizes at rest) = inducible ischemia (viable, jeopardized myocardium); a FIXED defect (deficit on both) = scar/prior infarct. Add imaging to ECG when the baseline ECG is uninterpretable.

BOARD TRAP

WRONG: a fixed defect represents active inducible ischemia needing revascularization. Discriminator: fixed = scar (non-viable, won't benefit from revascularization); only REVERSIBLE defects represent ischemia that revascularization can relieve.

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12. CMR transmural extent

WHAT YOU SEE

'CMR' heart images, 'TRANSMURAL EXTENT'

WHAT IT ENCODES

Cardiac MRI with late gadolinium enhancement grades the TRANSMURAL extent of scar to predict recovery after revascularization. Scar involving <50% of wall thickness is likely viable and will recover function; full-thickness (transmural) late enhancement = non-viable myocardium that will NOT recover after revascularization. CMR also provides high-resolution wall-motion and perfusion data.

BOARD TRAP

WRONG: transmural late-gadolinium scar will recover function once the vessel is reopened. Discriminator: transmural (>50% thickness) scar is non-viable and won't recover with revascularization — only segments with limited subendocardial scar (viable) improve.

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13. Cannot exercise → pharm injector

WHAT YOU SEE

Far-right patient in a wheelchair ('CANNOT EXERCISE') beside a pharmacologic-stress injector — can't run, so a drug delivers the stress instead.

WHAT IT ENCODES

Patients who cannot exercise adequately (deconditioning, arthritis, PAD, amputation, neurologic limits) get PHARMACOLOGIC stress instead: vasodilators (adenosine, dipyridamole, regadenoson) for PERFUSION imaging, or dobutamine for WALL-MOTION stress echo. Pharmacologic stress is preferred because a submaximal exercise test is non-diagnostic — you must achieve adequate stress to read the study.

BOARD TRAP

WRONG: a patient who can't exercise should still attempt a treadmill test and a submaximal result is acceptable. Discriminator: inability to reach target workload makes exercise ECG non-diagnostic — switch to pharmacologic stress rather than accepting an inadequate exercise study.
